

CS & IT ENGINEERING



Computer Network

CIDR Allocation

Lecture No. - 07



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Recap of Previous Lecture



Topic

Supernetting





Topics to be Covered



Topic

CIDR Allocation





Topic : Fixed Length Subnetting

→ Fixed length prefixing

[All subnets are equal in size]



Topic : Fixed Length Subnetting



→ All subnets uses same subnet mask.



Topic : Fixed Length Subnetting



→ Inefficient utilization of IP address space



Topic : Variable Length Subnetting



→ Variable Length Network prefixes

[Unlike fixed length prefixing]



Topic : Variable Length Subnetting



→ Allow variable size sub-networks (subnets)



Topic : Variable Length Subnetting



→ efficient utilization of IP address space



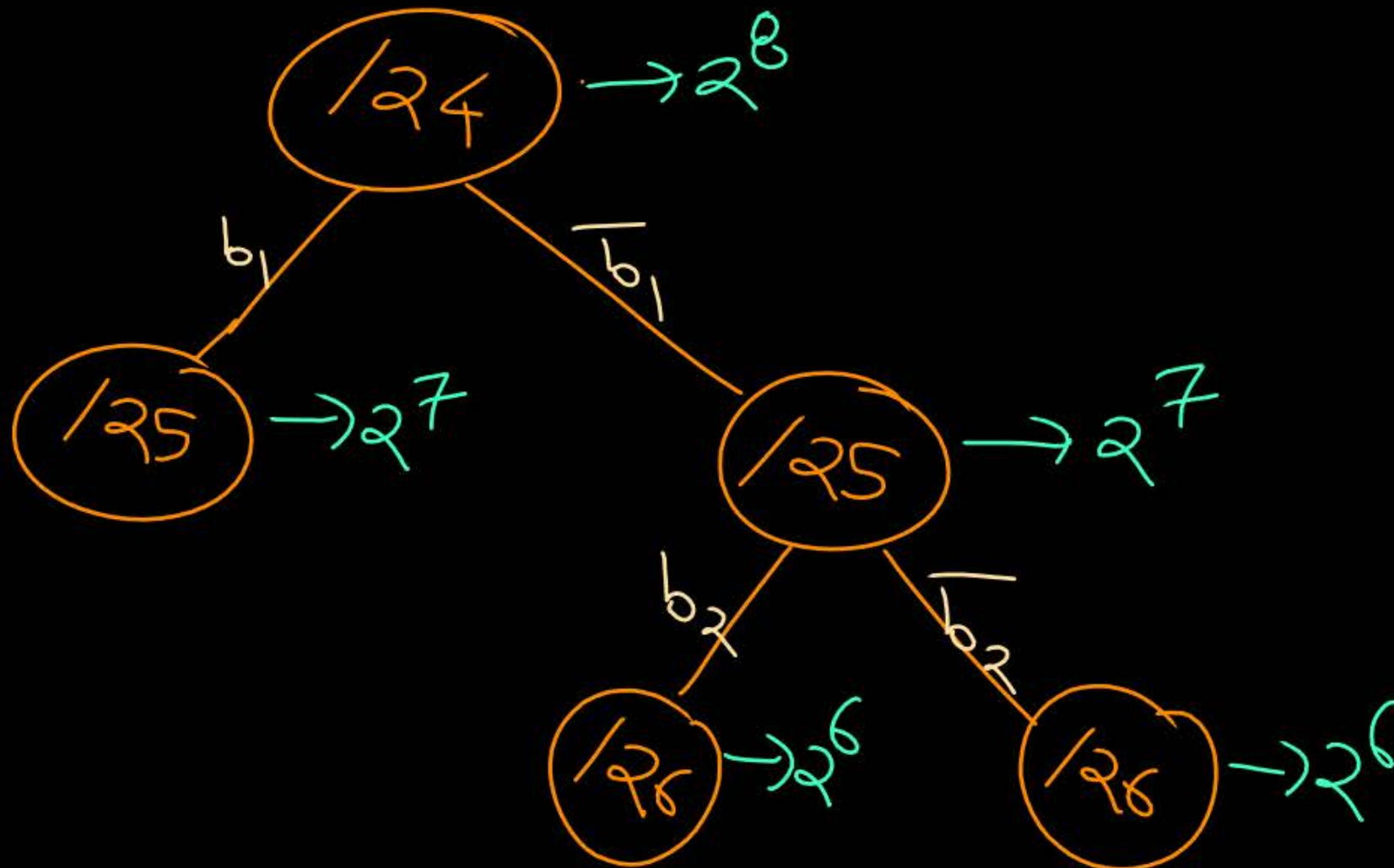
Topic : VLSM



- VLSM : Variable Length Subnet Mask
- Subnets may uses different subnet mask.

Example 7 :-

#Q. Consider a class-less IPv4 network, the network address of this network is '80 . 80 . 80 . 0 / 24'. The network is divided into three subnets, one subnet have approximately 50 percent hosts and two subnets each have approximately 25 percent hosts. Write subnet address address of all the three subnets?



VLSM → ^{IT} 2004/05

CIDR → CS-2012



Network Address : 80.80.80.0 / 24

Broadcast Address : 80.80.80.255 / 24

Network Address : 80.80.80.0 / 24 ✓

Broadcast Address : 80.80.80.255 / 24 ✓

Before Subnetting :-

First Host IP : 80.80.80.1 / 24

Last Host IP : 80.80.80.254 / 24

$$(2^8 - 2) = 254$$

Network Address : 80 . 80 . 80 . 0 / 24

Broadcast Address : 80 . 80 . 80 . 255 / 24

Before Subnetting :-

First Host IP : 80 . 80 . 80 . 1 / 24

Last Host IP : 80 . 80 . 80 . 254 / 24

Network Size = 254 Hosts

24 bit (NetID) Prefix $\rightarrow$ 8 bit hostID $\rightarrow 2^8$
80.80.80.b₁ b₂ /24



Subnet-1 Address
[for 50% hosts]

80.80.80.0 00000000 /25
25 bit prefix 7 bit hostID

$\rightarrow 2^7$

Subnet-2 Address
[for 25% hosts]

80.80.80.1 0 00000000 /26
26 bit prefix 6 bit hostID

$\rightarrow 2^6$

Subnet-3 Address
[for 25% hosts]

80.80.80.1 1 00000000 /26
26 bit prefix 6 bit hostID

$\rightarrow 2^6$

Subnet-1 Address : 80 . 80 . 80 . 0 / 25

Subnet-2 Address : 80 . 80 . 80 . 128 / 26

Subnet-3 Address : 80 . 80 . 80 . 192 / 26

Subnet-1 Address : 80 . 80 . 80 . 0 / 25

Subnet-2 Address : 80 . 80 . 80 . 192 / 26

Subnet-3 Address : 80 . 80 . 80 . 128 / 26

24 bit (NetID) Prefix $\rightarrow$ 8 bit hostID $\rightarrow 2^8$
80.80.80.b₁ b₂ ----- / 24



Subnet-1 Address
[for 50% hosts]

80.80.80.1 00000000 / 25 $\rightarrow 2^7$
25 bit prefix 7 bit hostID

Subnet-2 Address
[for 25% hosts]

80.80.80.00 00000000 / 26 $\rightarrow 2^6$
26 bit prefix 6 bit hostID

Subnet-3 Address
[for 25% hosts]

80.80.80.01 00000000 / 26 $\rightarrow 2^6$
26 bit prefix 6 bit hostID

Subnet-1 Address : 80 . 80 . 80 . 128 / 25

Subnet-2 Address : 80 . 80 . 80 . 0 / 26

Subnet-3 Address : 80 . 80 . 80 . 64 / 26

Subnet-1 Address : 80 . 80 . 80 . 128 / 25

Subnet-2 Address : 80 . 80 . 80 . 64 / 26

Subnet-3 Address : 80 . 80 . 80 . 0 / 26



Topic : CIDR



→ IP Address allocation method for IP routing



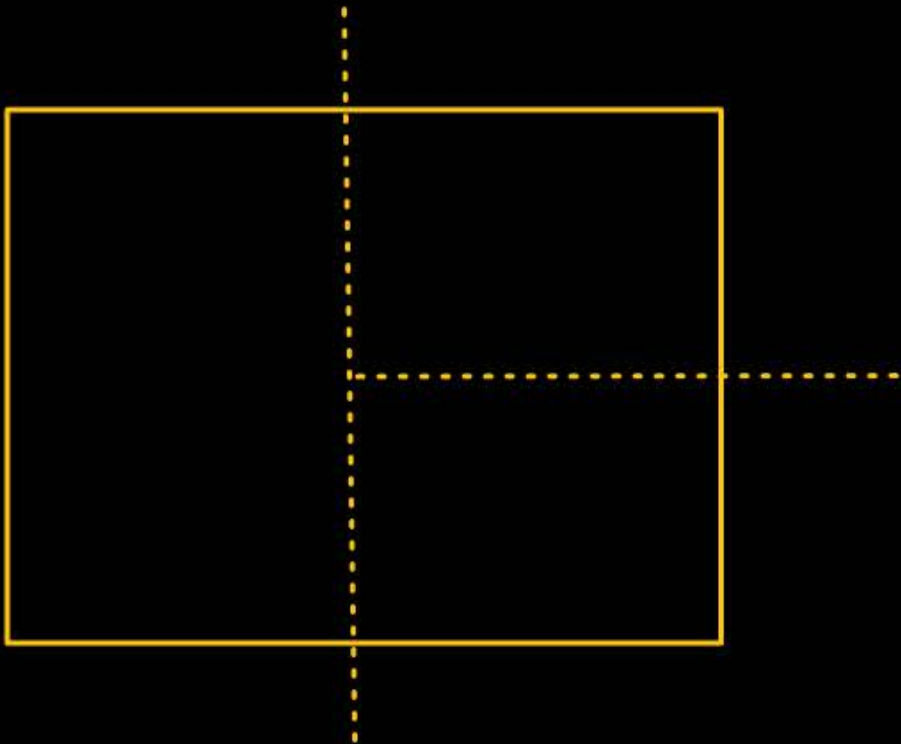
Topic : CIDR



→ Based on Variable-length subnet masking (VLSM)

Example 8 :-

#Q. An Internet Service Provider (ISP) has the following chunk of CIDR-based IP addresses available with it : "205 . 207 . 209 . 0 / 24". An organization request to ISP for range of IP address for its 45 hosts. Write all the possible CIDR prefixes which can be used for this allocation?



CS-20
CS-26

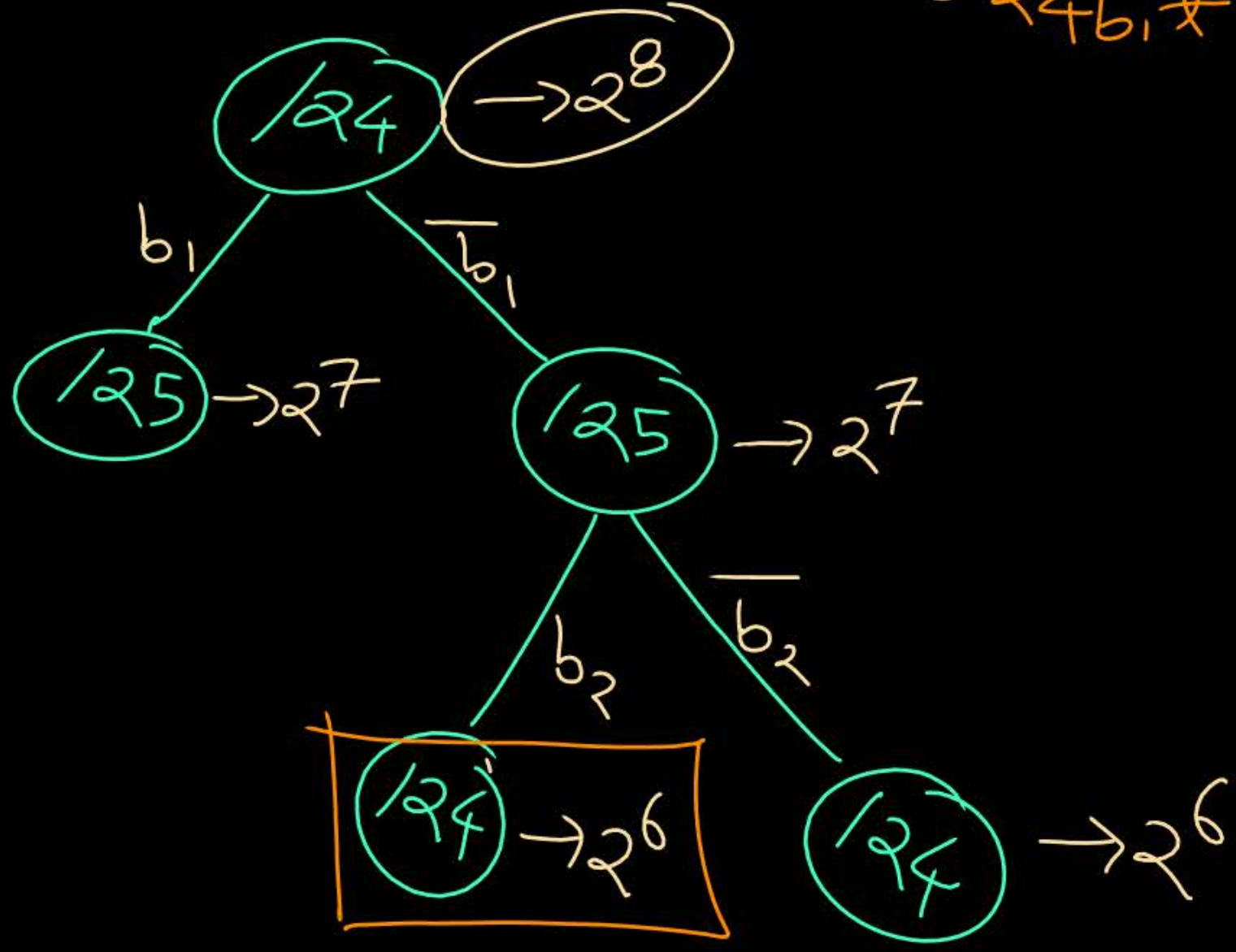


Prefix (24 bits) $\rightarrow$ 8 bits $(\rightarrow 2^8 \text{ IPs})$

Available IP Address space = 205.207.209.0 / 24

205.207.209. $\underline{b_1} \underline{b_2} \dots \dots \dots$ / 24

24 bit



$$\underbrace{\text{Number of hosts}}_{\text{[for IP address requested]}} = \underline{45}$$

$$\boxed{\leftarrow 2^6 \text{ IP chunk}}$$

$$\begin{aligned}\underbrace{\text{Number of bits for host id in allocated chunk}} &= \lceil \log_2(\text{Number of hosts}) \rceil \text{ bits} \\ &= \lceil \log_2(45) \rceil \text{ bits} \\ &= 6 \text{ bits}\end{aligned}$$

↓
 Allocated IP Address space = 205.207.209. PQ 00000000 / 26
 26 bit prefix 6 bit host ID

PQ → $b_1, b_2, \overline{b_1}, \overline{b_2}, \overline{b_1}, \overline{b_2}$

00, 01, 10, 11

CIDR Prefix = 205.207.209. 0 / 26

OR

= 205.207.209. 64 / 26

OR

= 205.207.209. 128 / 26

OR

= 205.207.209. 192 / 26



2 mins Summary



Topic

CIDR Allocation ✓



THANK - YOU